The Server APIs

PHP was originally designed to run embedded in web pages, providing server-side processing before the server sent a response. PHP is now over two decades old and has long since grown into a richly featured general-purpose language that can be used for a wide variety of system-level tasks. An intrinsic aspect of PHP's execution is the Server API (SAPI). The SAPI governs the communication between your program running within the PHP engine and everything else happening outside your code, and there are different SAPIs tuned for the general requirements of the environment in which PHP is executing.

A PHP program that is called from a command shell has different needs to one that is servicing an HTTP request. A shell command almost never needs to worry about HTTP headers but might need to interact with a command shell's environment or obtain user input supplied when the program was invoked. Conversely, a web server might set a maximum execution time (there's no point in continuing to run code that has exceeded the timeout of the HTTP request) and restrict which parts of the filesystem PHP can access as a security precaution.

To determine at run-time which SAPI your code is working with, you can either call `php_sapi_name()` or find the value of the built-in `PHP_SAPI` constant:

```
$ php -a
Interactive shell

php > echo php_sapi_name();
cli
php > echo PHP_SAPI;
cli
php >
```

The major differences are between the `cli` SAPI and all the others. Every non-CLI SAPI will be running as some form of web-server (such as `php-fpm`, `mod_apache`) and needs to deal with the overheads of the HTTP protocol and standards for data transfer (such as email attachments): we'll cover these things in detail in Chapter 11, "Programming for the Internet." The `cli` SAPI differs from web SAPIs in the following `.ini` directives: